

Problem 26.15

An annuity pays 10 at the end of year 2, and 9 at the end of year 4. The payments continue decreasing by 1 each two year period until 1 is paid at the end of year 20. Calculate the present value of the annuity at an annual effective interest rate of 5%.

Solution

The payments occur every two years:

$$10, 9, 8, \dots, 1.$$

Therefore, we use two-year periods. If j is the effective interest rate per two-year period, then

$$1 + j = (1.05)^2.$$

Thus,

$$j = (1.05)^2 - 1 = 0.1025.$$

The payment stream is a decreasing annuity-immediate with 10 payments, so its present value is

$$PV = (Da)_{\overline{10}|j}.$$

For a decreasing annuity-immediate,

$$(Da)_{\overline{n}|} = \frac{n - a_{\overline{n}|}}{i}.$$

Using the two-year rate j ,

$$PV = \frac{10 - a_{\overline{10}|j}}{j}.$$

Now,

$$a_{\overline{10}|j} = \frac{1 - (1 + j)^{-10}}{j} = \frac{1 - (1.1025)^{-10}}{0.1025} = 6.079126996.$$

Therefore,

$$PV = \frac{10 - 6.079126996}{0.1025} = 38.25241955.$$

Hence, the present value is

$$\boxed{38.25}.$$